

**궁내교
수위 예측 AI
성능 개선 결과 3**
(시계열)

※ 강우 이벤트 전부 학습으로 사용

MIET 학습 데이터 기본 설정

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- learning_rate = 0.0001
- Loss_function = MSE
- Seq_length = 36 (6시간)
- Num_of_layer = 12 (각 LSTM 3개 처리)

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 : 총 1420개 中 1412개 데이터 파일 학습데이터로 사용 (결측 데이터 5개)

검증 : 학습 데이터에서 제외한 3개 데이터 사용 ++ 학습에서 검증에 사용된 3개 데이터 사용

- 학습 : ~ 2022 68번 (988개)
- 검증 : ~ 2024 96번 (282개)
- 테스트 : ~ 2025 109번 (142개)

seq_length 조정

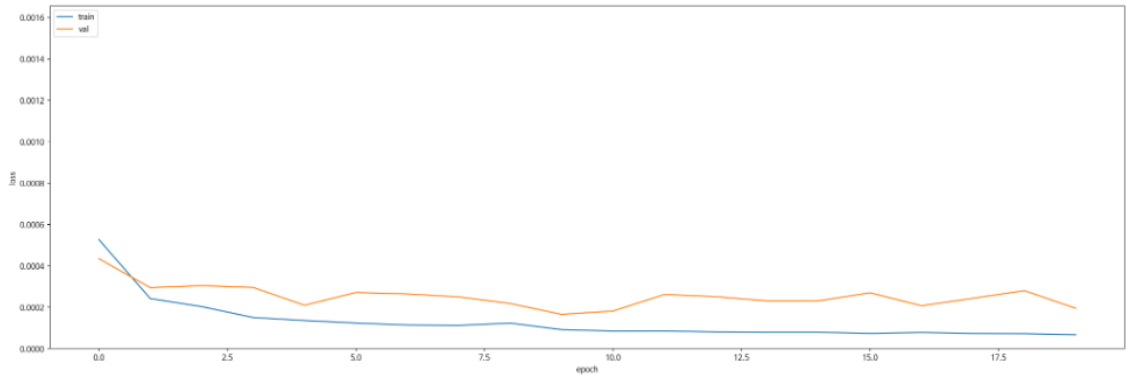
0. seq_length = 36

seq_length = 36

학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

Train (23111 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0258	0.0245	0.0383	0.967146	0.969125	6.806384

Valid (6828 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0371	0.0399	0.0522	0.946929	0.953082	8.452243

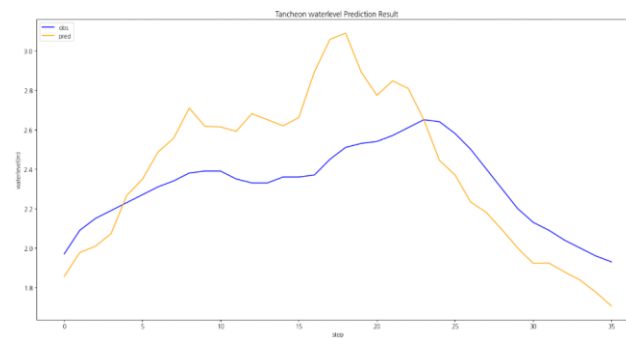
Test (5472 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0416	0.0481	0.0583	0.904372	0.927073	2.905314

seq_length = 36
예측 코드 - 궁내교

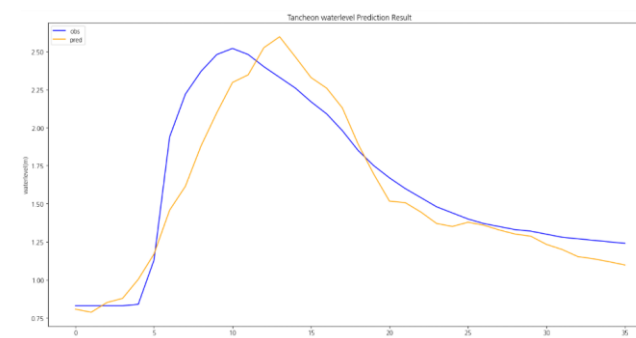
궁내교	
testdata_220712	r2 : -0.7945404436174226 mse : 0.07120448467610142 mae : 0.23399888343281214 rmse : 0.26684168466733493
testdata_230711	r2 : 0.857466599365075 mse : 0.03988668540781691 mae : 0.14282773759629994 rmse : 0.199716512606787
증설 후	
testdata_250716	r2 : 0.8885925911762368 mse : 0.011176638825219534 mae : 0.07951597525013816 rmse : 0.1057196236524683
testdata_250814	r2 : 0.7262170094360034 mse : 0.019302271215992088 mae : 0.11494690630171034 rmse : 0.13893261393924786

증설 전

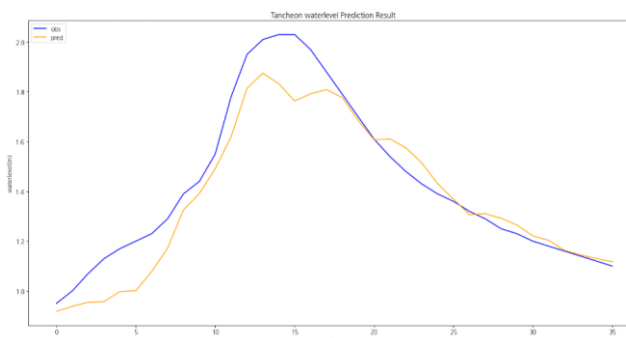
증설 후



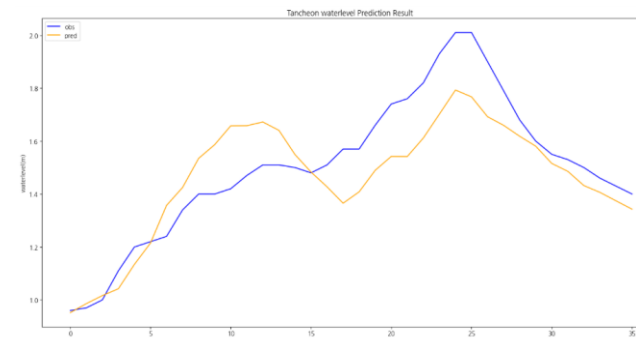
testdata_220712



testdata_230711



testdata_250716



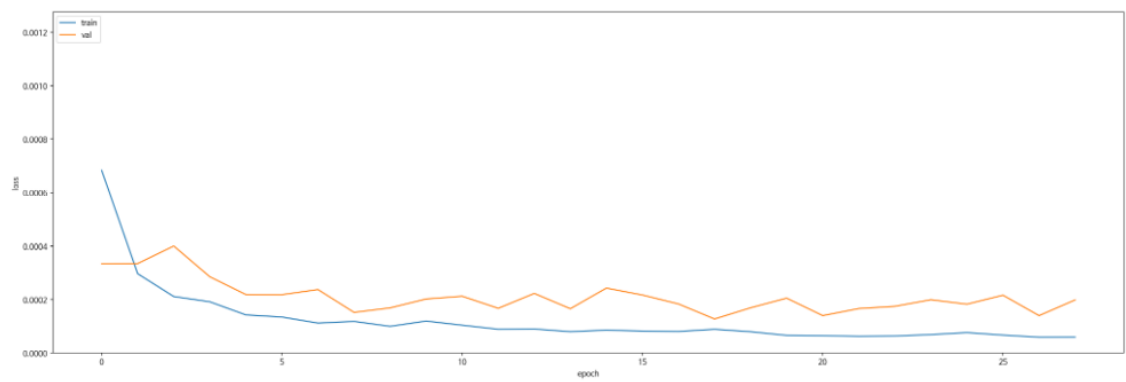
testdata_250814

1. seq_length = 48

seq_length = 48
학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

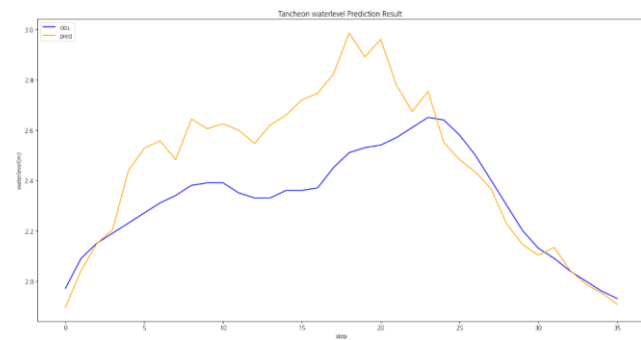
Train (13486 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0242	0.0233	0.0358	0.972953	0.976071	1.410352

Valid (3589 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0333	0.0363	0.0458	0.958829	0.962585	12.163913

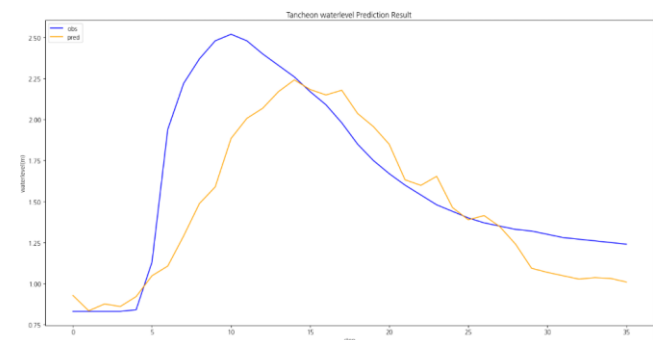
Test (3768 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0390	0.0451	0.0551	0.916871	0.933745	1.162779

seq_length = 48
예측 코드 - 궁내교

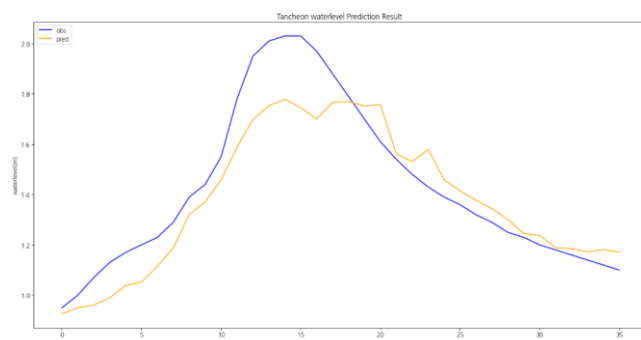
	궁내교
증설 전	testdata_220712 r2 : -0.1823864103244639 mse : 0.04691519510447297 mae : 0.16771526826752556 rmse : 0.21659915767258417
	testdata_230711 r2 : 0.554343110493201 mse : 0.12471305723711411 mae : 0.23296182599332596 rmse : 0.353147359096899
증설 후	testdata_250716 r2 : 0.8364645915436455 mse : 0.016406235588360272 mae : 0.1013863165511025 rmse : 0.12808682831720158
	testdata_250814 r2 : 0.5842686132385491 mse : 0.02930992887373803 mae : 0.14634109947416515 rmse : 0.17120142777949612



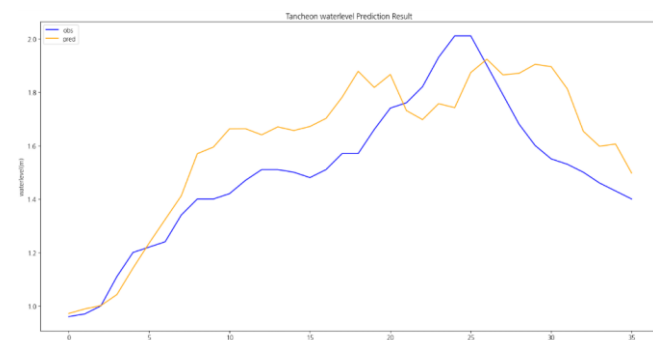
testdata_220712



testdata_230711



testdata_250716



testdata_250814

기존 seq_length = 36 이 제일 나음

seq_length = 36
+ num_of_layer 조정

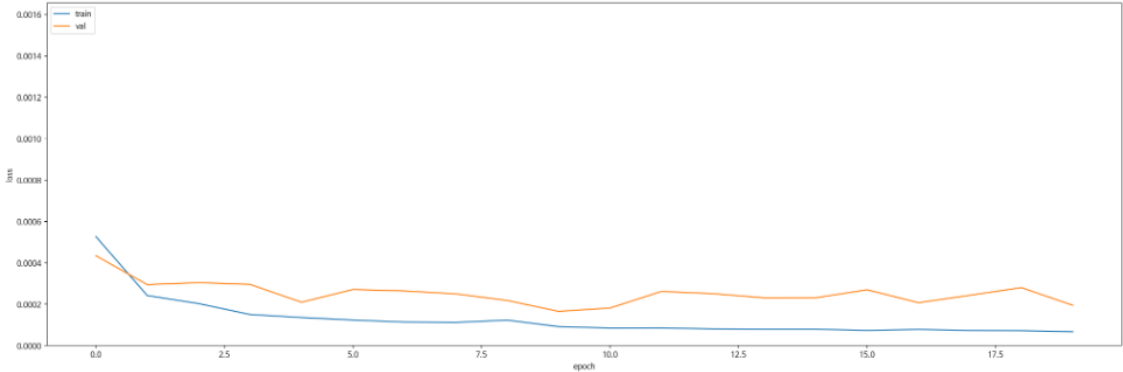
0. num_of_layer = 12

num_of_layer =12

학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

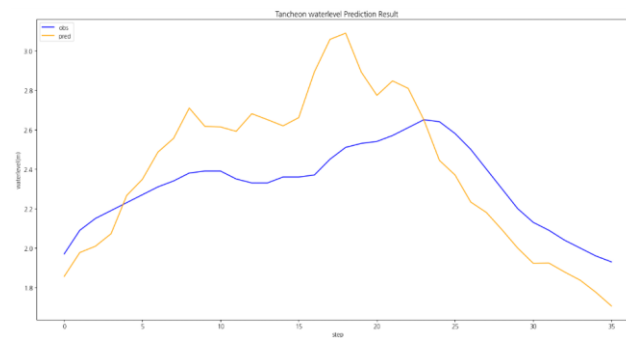
Train (23111 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0258	0.0245	0.0383	0.967146	0.969125	6.806384

Valid (6828 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0371	0.0399	0.0522	0.946929	0.953082	8.452243

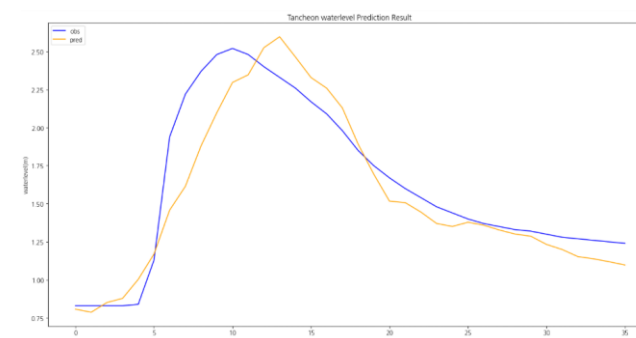
Test (5472 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0416	0.0481	0.0583	0.904372	0.927073	2.905314

num_of_layer =12
예측 코드 - 궁내교

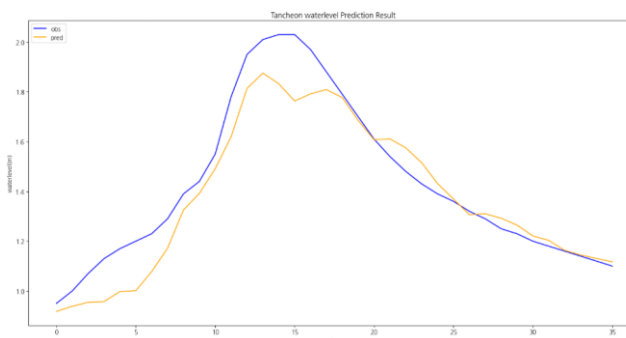
	궁내교
증설 전	testdata_220712 r2 : -0.7945404436174226 mse : 0.07120448467610142 mae : 0.23399888343281214 rmse : 0.26684168466733493
	testdata_230711 r2 : 0.857466599365075 mse : 0.03988668540781691 mae : 0.14282773759629994 rmse : 0.199716512606787
증설 후	testdata_250716 r2 : 0.8885925911762368 mse : 0.011176638825219534 mae : 0.07951597525013816 rmse : 0.1057196236524683
	testdata_250814 r2 : 0.7262170094360034 mse : 0.019302271215992088 mae : 0.11494690630171034 rmse : 0.13893261393924786



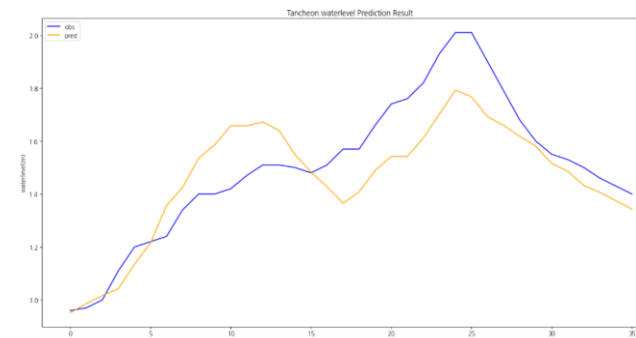
testdata_220712



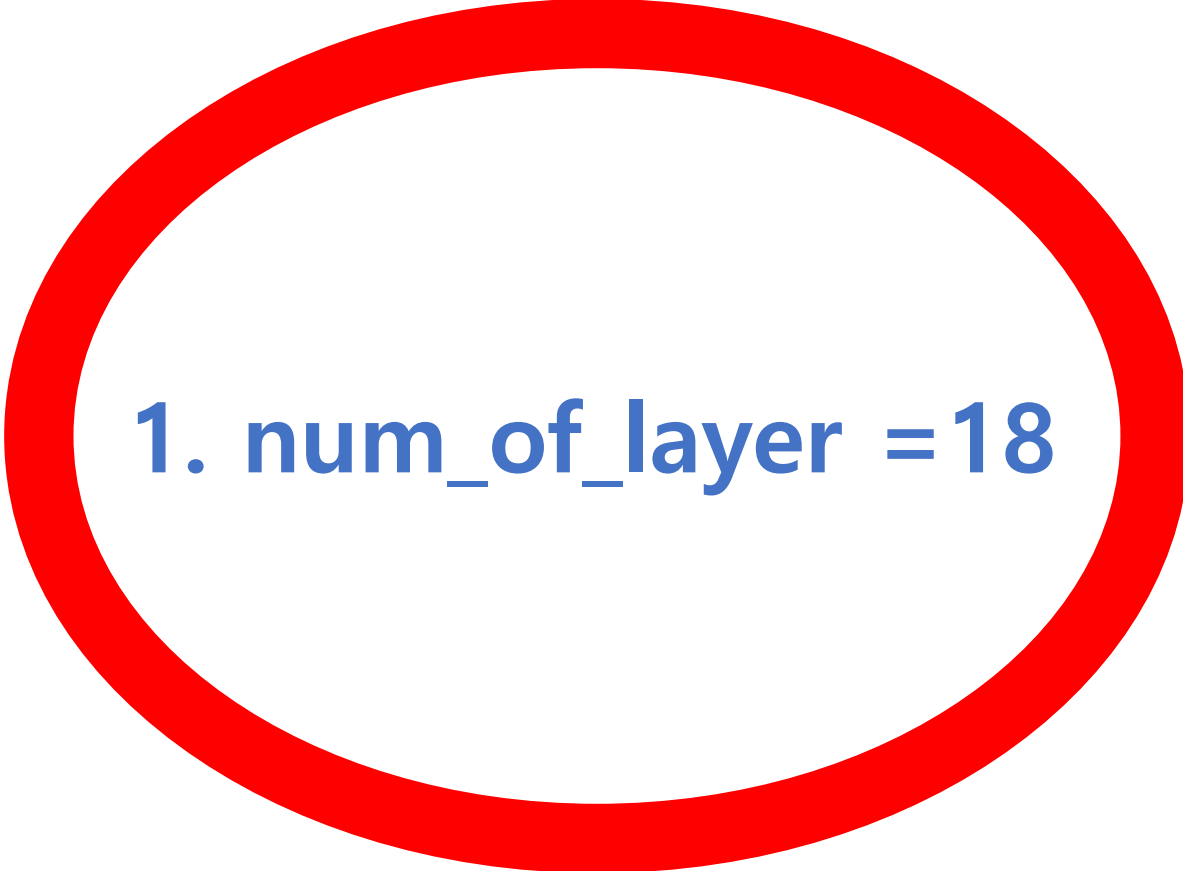
testdata_230711



testdata_250716



testdata_250814

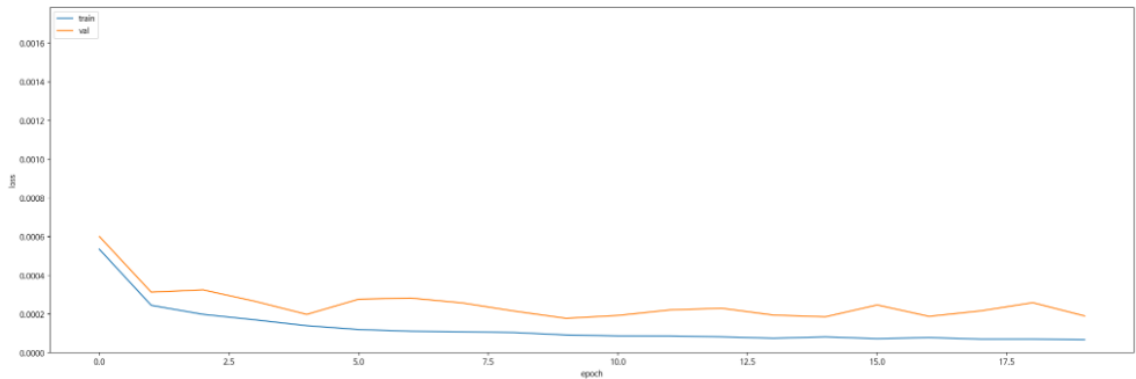


1. num_of_layer = 18

num_of_layer =18
학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

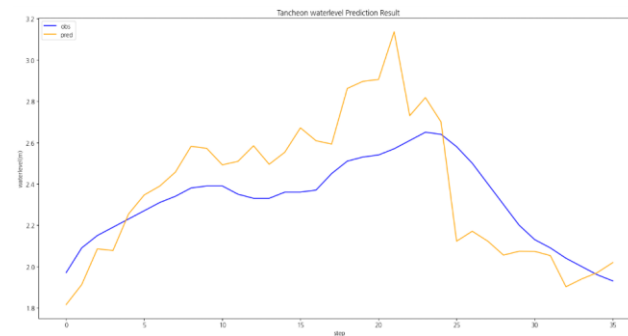
Train (23111 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0254	0.0242	0.0372	0.968928	0.970272	22.160994

Valid (6282 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0385	0.0421	0.0542	0.942801	0.951583	4.037687

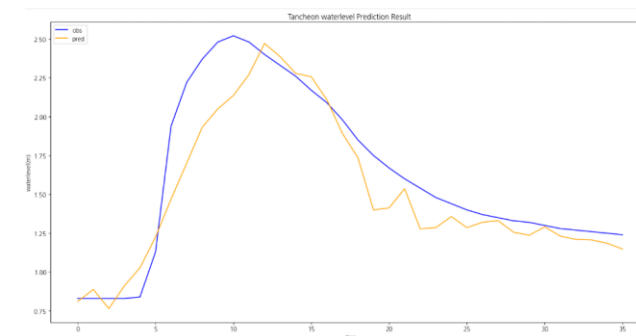
Test (5472 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0453	0.0528	0.0625	0.890020	0.919186	5.456180

num_of_layer =18
예측 코드 - 궁내교

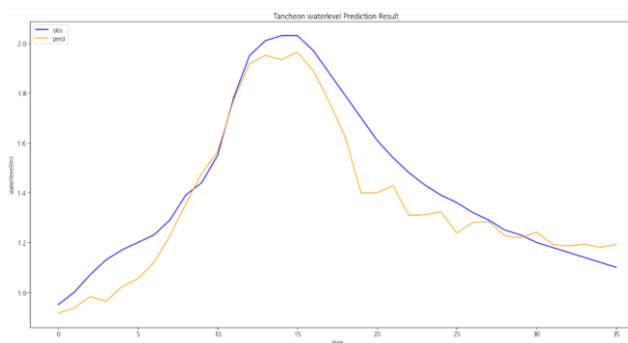
	궁내교
증설 전	testdata_220712 r2 : -0.2437545822983267 mse : 0.04935018567626799 mae : 0.18286159541871813 rmse : 0.22214901682489616
	testdata_230711 r2 : 0.8500162646896093 mse : 0.04197159430678016 mae : 0.14689570089181267 rmse : 0.2048697008021932
증설 후	testdata_250716 r2 : 0.8895288897883276 mse : 0.011082707267791009 mae : 0.08337979230615827 rmse : 0.1052744378649965
	testdata_250814 r2 : 0.7838021043271743 mse : 0.015242402057216858 mae : 0.0962281313869688 rmse : 0.1234601233484596



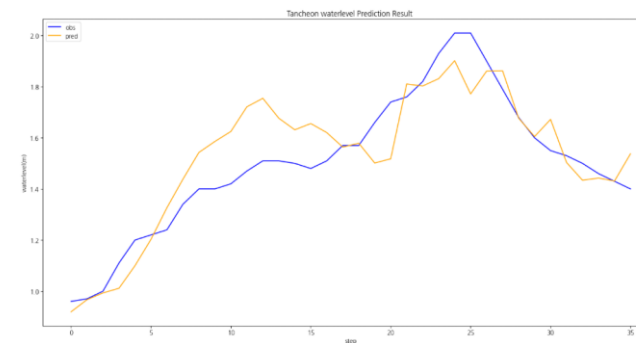
testdata_220712



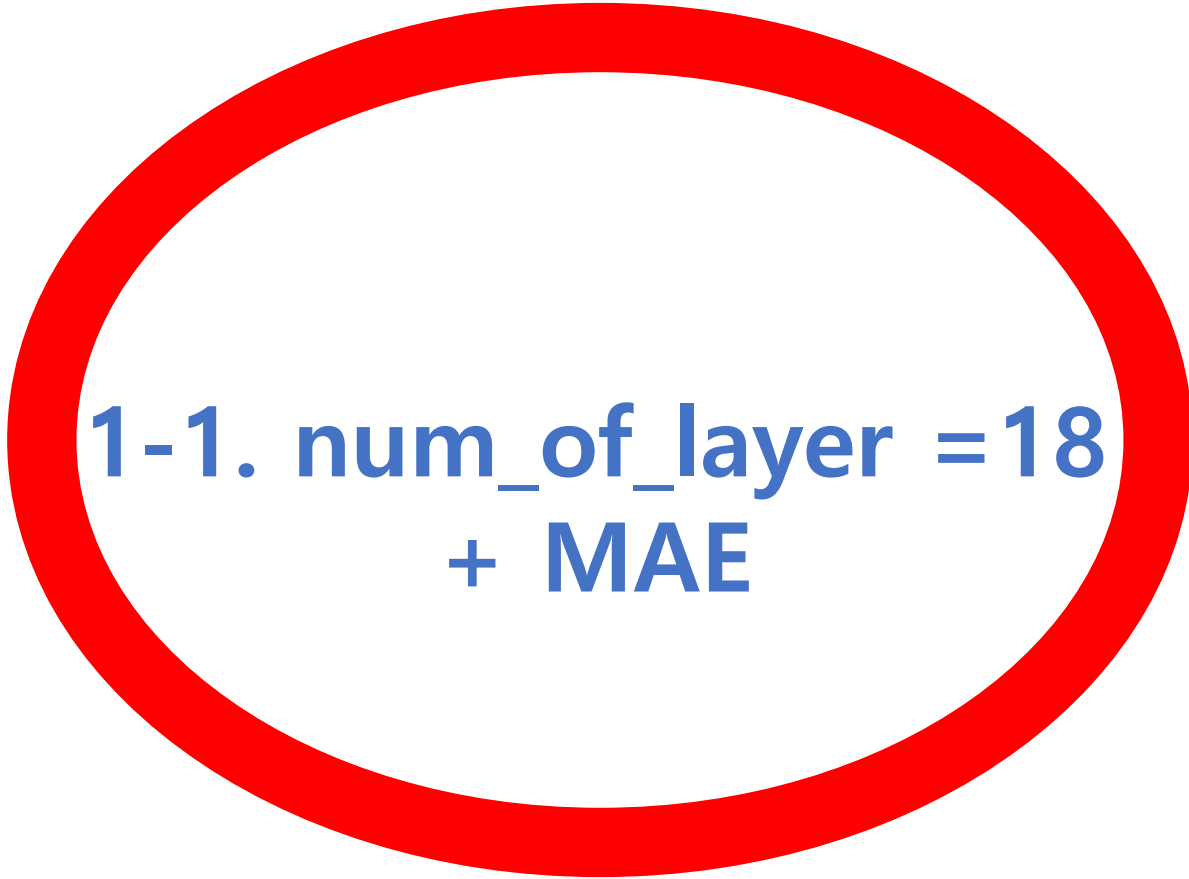
testdata_230711



testdata_250716



testdata_250814

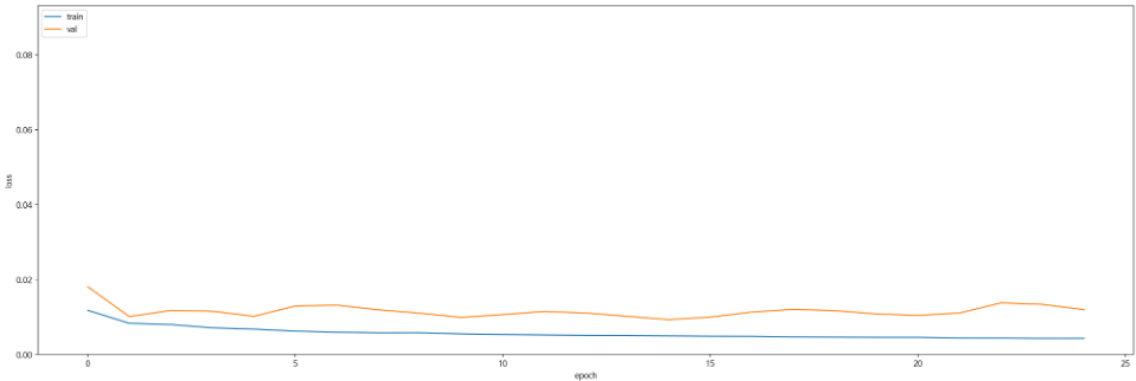


**1-1. num_of_layer = 18
+ MAE**

num_of_layer =18
loss_function = MAE
학습 코드 - 공내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

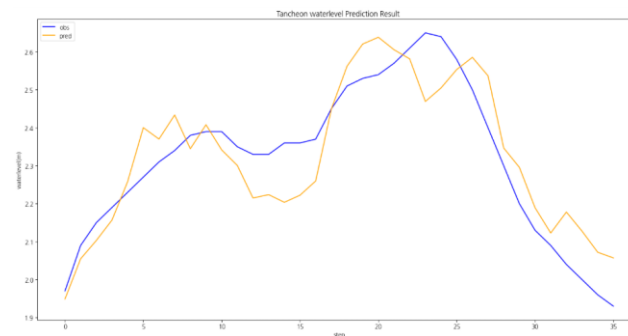
Train (23111 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0206	0.0197	0.0312	0.978130	0.980108	12.456093

Valid (6282 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0378	0.0424	0.0551	0.940861	0.952862	3.006159

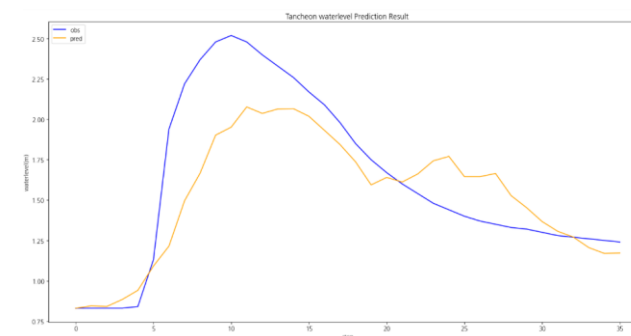
Test (5472 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0453	0.0538	0.0642	0.884159	0.915125	4.980753

num_of_layer = 18
 loss_function = MAE
 예측 코드 - 궁내교

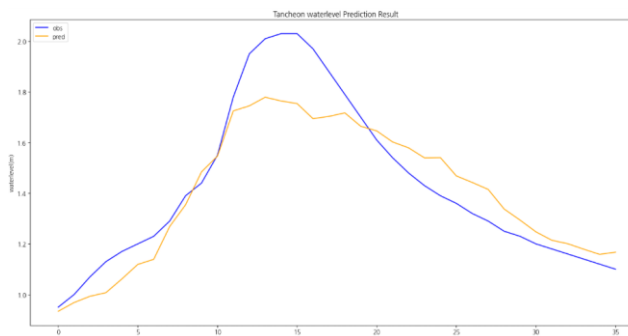
	궁내교
증설 전	testdata_220712 r2 : 0.7886506277757681 mse : 0.00838600388716136 mae : 0.07876580847634213 rmse : 0.09157512701143995
	testdata_230711 r2 : 0.6794901382392365 mse : 0.0896917913219421 mae : 0.2132765921619203 rmse : 0.29948587833475904
증설 후	testdata_250716 r2 : 0.8537863174988561 mse : 0.014668481547809194 mae : 0.09608093248473272 rmse : 0.12111350687602598
	testdata_250814 r2 : 0.7326166026610431 mse : 0.01885108656114091 mae : 0.09745151480038959 rmse : 0.13729925914272412



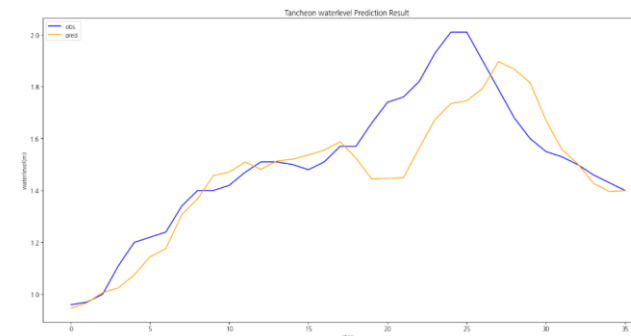
testdata_220712



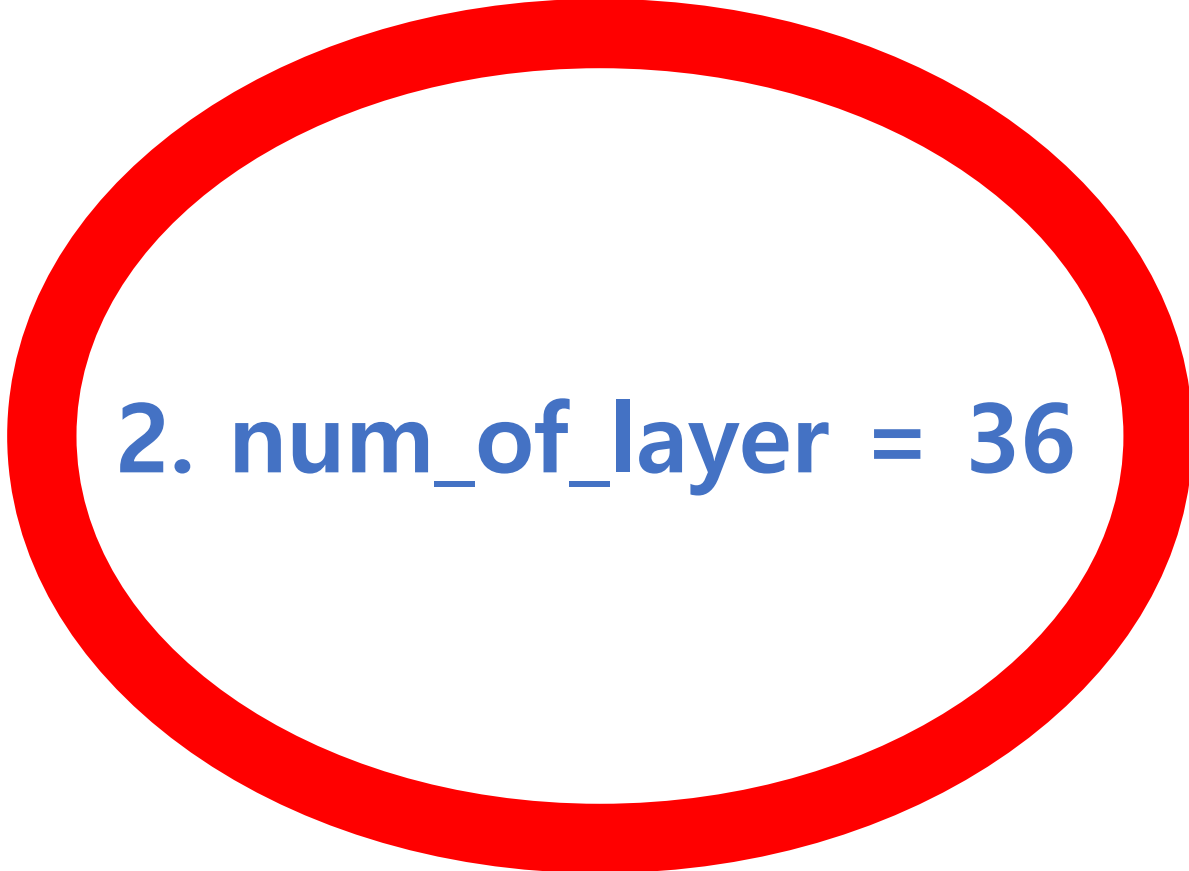
testdata_230711



testdata_250716



testdata_250814

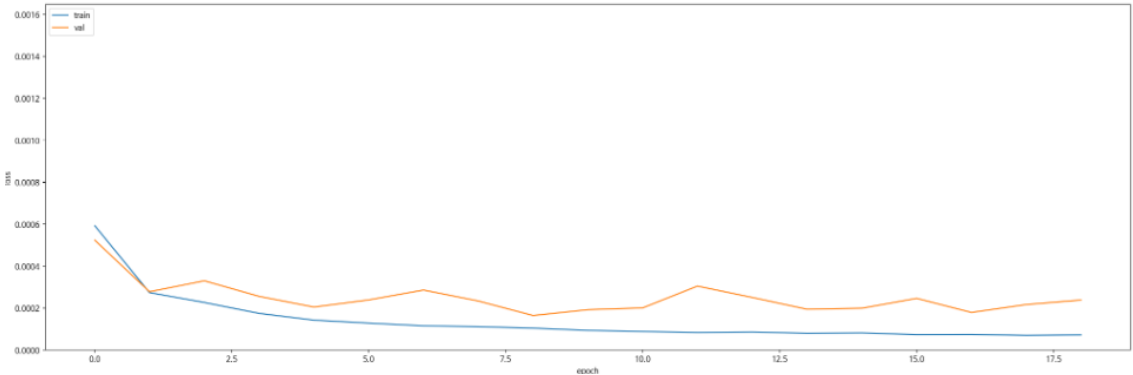


2. num_of_layer = 36

num_of_layer =36
학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

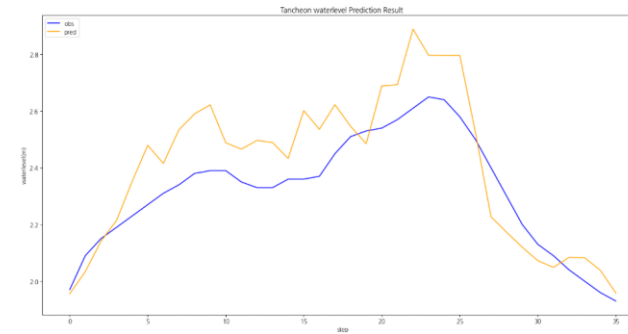
Train (23111 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0269	0.0256	0.0388	0.966277	0.970299	1.513945

Valid (6822 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0378	0.0409	0.0521	0.947172	0.956457	1.215834

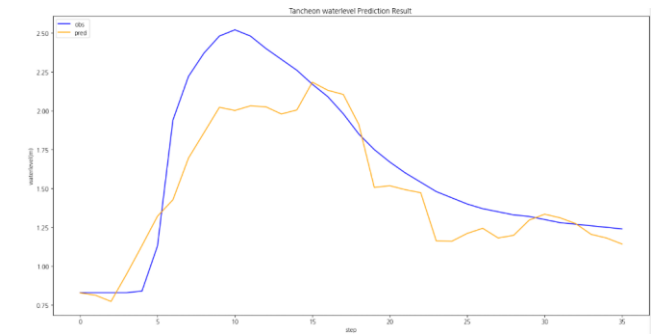
Test (5472 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0407	0.0468	0.0569	0.908998	0.930175	2.279257

num_of_layer =36
예측 코드 - 궁내교

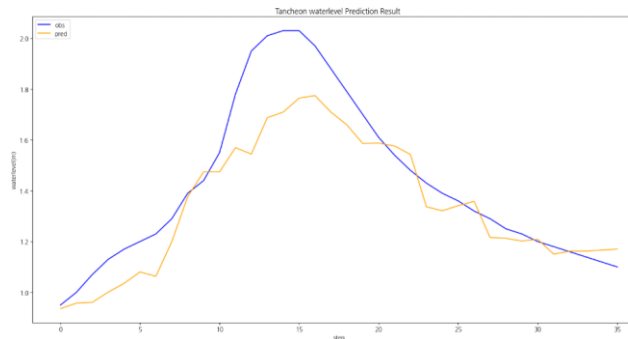
	궁내교
증설 전	testdata_220712 r2 : 0.5154125887431622 mse : 0.01922765074578906 mae : 0.11833648959795633 rmse : 0.13866380474294315
	testdata_230711 r2 : 0.7651658869889291 mse : 0.06571620649596127 mae : 0.19366505410936144 rmse : 0.2563517241915125
증설 후	testdata_250716 r2 : 0.7987440252756075 mse : 0.020190446619850436 mae : 0.10343127078480188 rmse : 0.14209309138677514
	testdata_250814 r2 : 0.9320650485021252 mse : 0.004789555611749123 mae : 0.05961067683166928 rmse : 0.06920661537562087



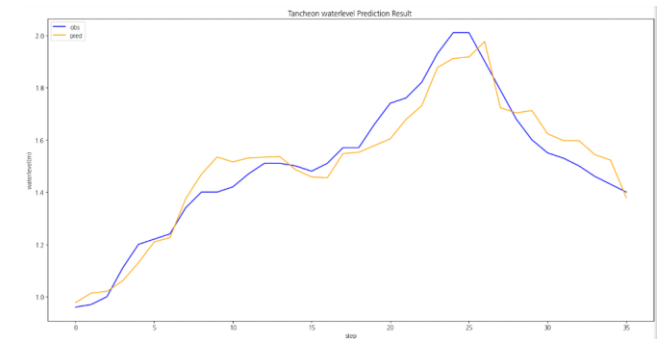
testdata_220712



testdata_230711



testdata_250716



testdata_250814

**seq_length = 36 +
num_of_layer = 18 이 제일 나옴**

Loss_function = MAE
Learning_rate = 0.0005

1. num_of_layer = 12

MIET 학습 데이터 기본 설정

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- learning_rate = 0.0005
- Loss_function = MAE
- Seq_length = 36 (6시간)
- Num_of_layer = 12 (각 LSTM 3개 처리)

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 : 총 1420개 中 1412개 데이터 파일 학습데이터로 사용 (결측 데이터 5개)

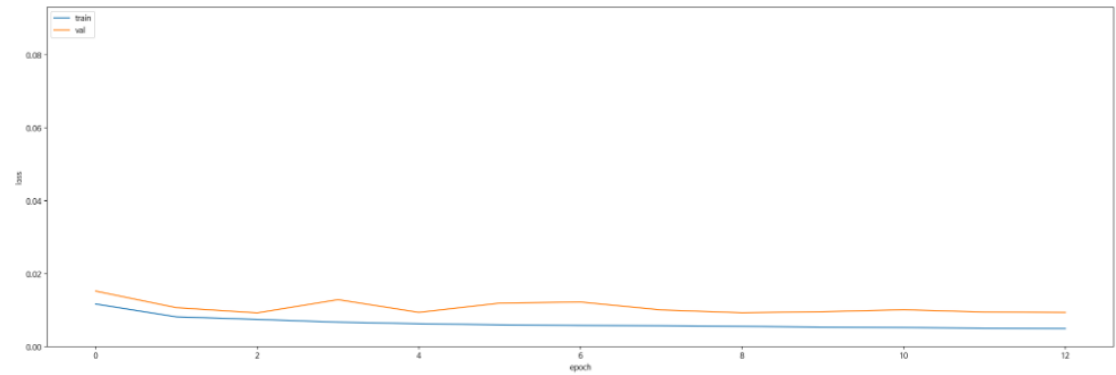
검증 : 학습 데이터에서 제외한 3개 데이터 사용 ++ 학습에서 검증에 사용된 3개 데이터 사용

- 학습 : ~ 2022 68번 (988개)
- 검증 : ~ 2024 96번 (282개)
- 테스트 : ~ 2025 109번 (142개)

num_of_layer = 12
학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

Train (23111 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0288	0.0264	0.0537	0.935251	0.936525	21.145243

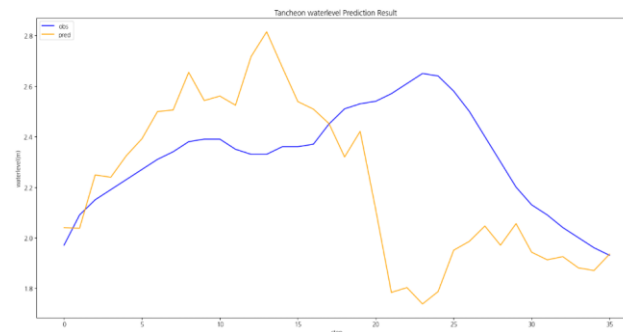
Valid (6828 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0377	0.0406	0.0585	0.933385	0.939242	3.664063

Test (5472 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0448	0.0509	0.0672	0.873095	0.891142	7.571774

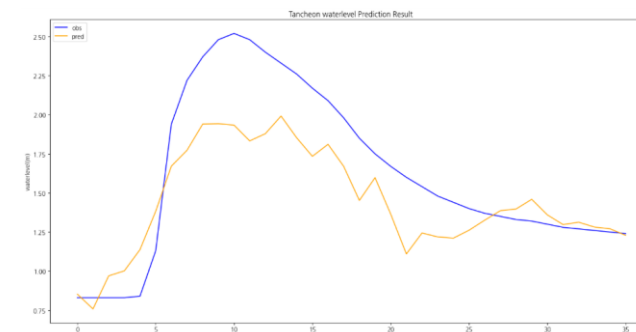
num_of_layer = 12

예측 코드 - 궁내교

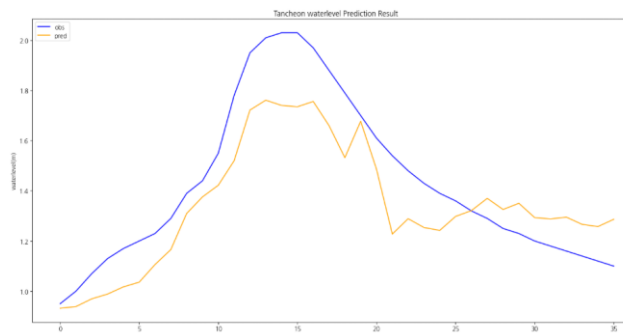
	궁내교
증설 전	testdata_220712 r2 : -2.419031330681882 mse : 0.13566167586722264 mae : 0.27432608577940193 rmse : 0.3683227876024271
	testdata_230711 r2 : 0.6617230110332062 mse : 0.09466376147287353 mae : 0.24711518228054047 rmse : 0.3076747657395282
증설 후	testdata_250716 r2 : 0.7240690801943894 mse : 0.02768200305472064 mae : 0.14641764938831325 rmse : 0.1663790944040766
	testdata_250814 r2 : 0.08522785843810254 mse : 0.06449334175540868 mae : 0.18133603036403656 rmse : 0.2539553932394598



testdata_220712



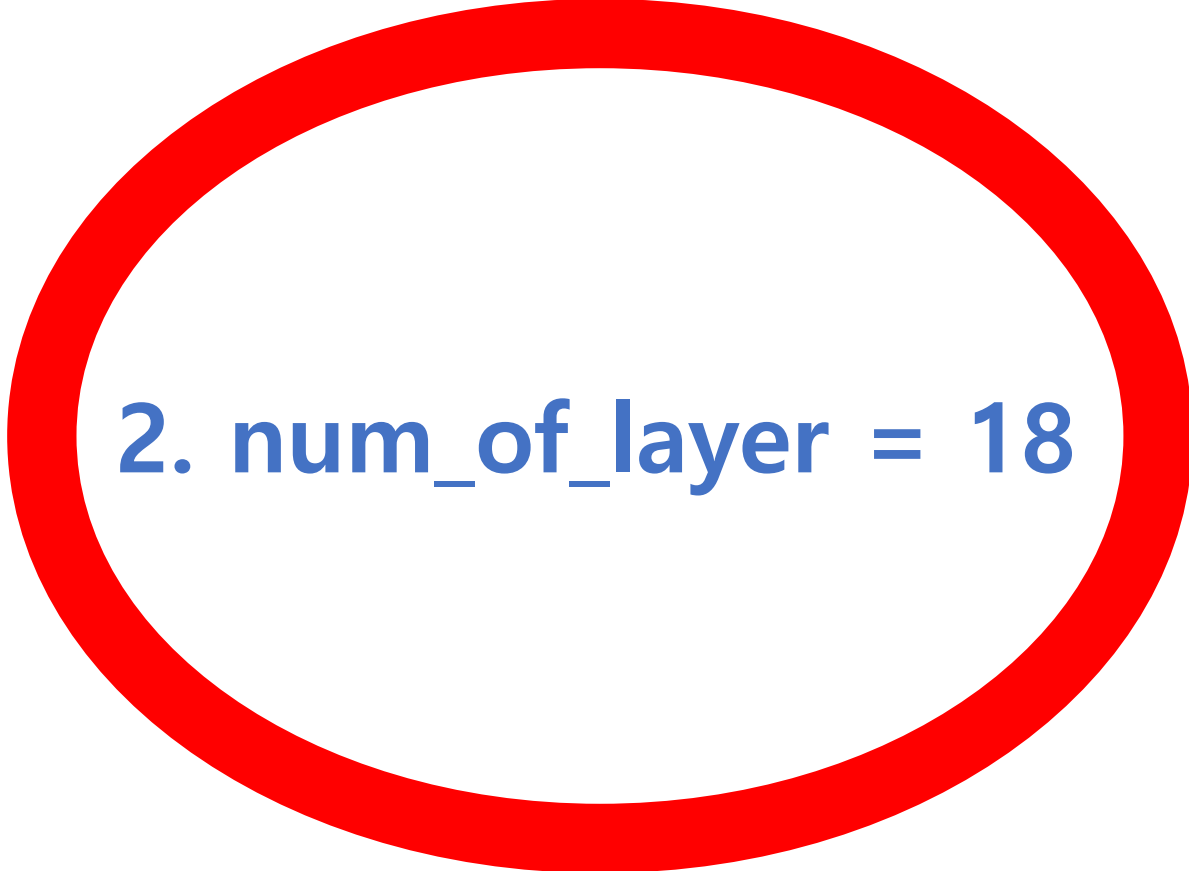
testdata_230711



testdata_250716



testdata_250814



2. num_of_layer = 18

MIET 학습 데이터 기본 설정

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- learning_rate = 0.0005
- Loss_function = MAE
- Seq_length = 36 (6시간)
- Num_of_layer = 18 (각 LSTM 3개 처리)

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 : 총 1420개 中 1412개 데이터 파일 학습데이터로 사용 (결측 데이터 5개)

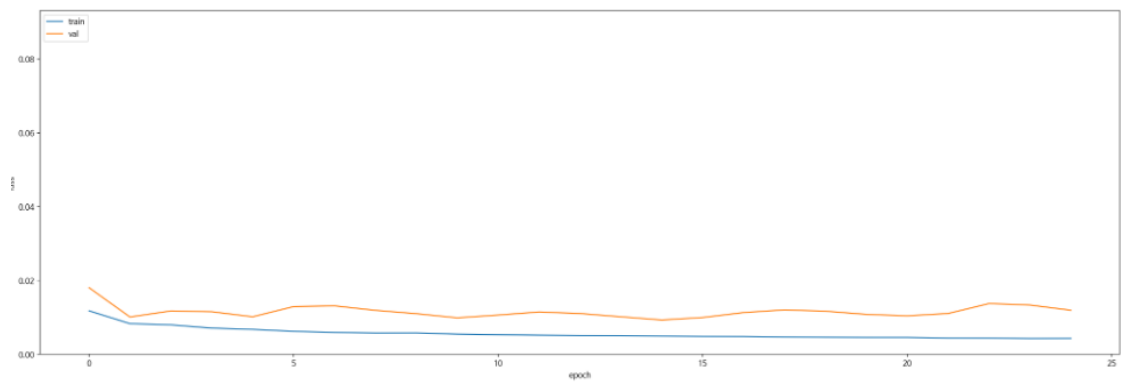
검증 : 학습 데이터에서 제외한 3개 데이터 사용 ++ 학습에서 검증에 사용된 3개 데이터 사용

- 학습 : ~ 2022 68번 (988개)
- 검증 : ~ 2024 96번 (282개)
- 테스트 : ~ 2025 109번 (142개)

num_of_layer = 18
학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

Train (23111 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0206	0.0197	0.0312	0.978130	0.980108	12.456093

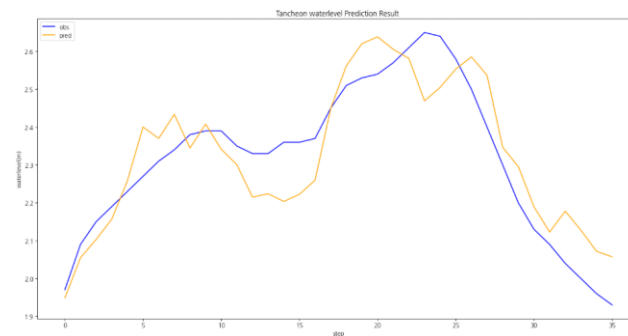
Valid (6828 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0378	0.0424	0.0551	0.940861	0.952862	3.006159

Test (5472 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0453	0.0538	0.0642	0.884159	0.915125	4.980753

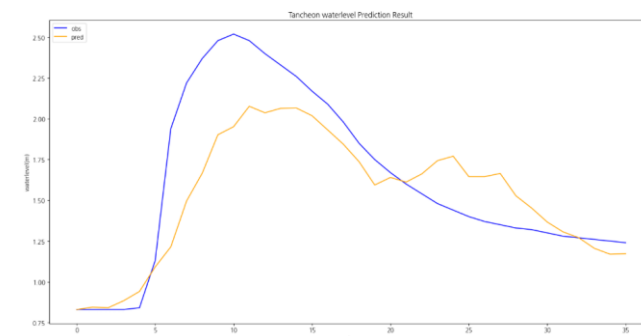
num_of_layer = 18

예측 코드 - 궁내교

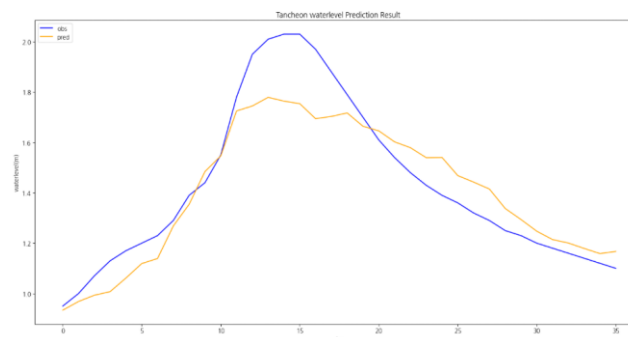
	궁내교
증설 전	testdata_220712 r2 : 0.7886506277757681 mse : 0.00838600388716136 mae : 0.07876580847634213 rmse : 0.09157512701143995
	testdata_230711 r2 : 0.6794901382392365 mse : 0.0896917913219421 mae : 0.2132765921619203 rmse : 0.29948587833475904
증설 후	testdata_250716 r2 : 0.8537863174988561 mse : 0.014668481547809194 mae : 0.09608093248473272 rmse : 0.12111350687602598
	testdata_250814 r2 : 0.7326166026610431 mse : 0.01885108656114091 mae : 0.09745151480038959 rmse : 0.13729925914272412



testdata_220712



testdata_230711



testdata_250716



testdata_250814

Loss_function = MAE
Learning_rate = 0.0005 도
num_of_layer = 18 이 제일 나옴

영향 없는 관측소 및 컬럼 빼기
+ seq_length 조정

MIET 학습 데이터 기본 설정

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- learning_rate = 0.0001
- Loss_function = MSE
- Seq_length = 36 (6시간)
- Num_of_layer = 12 (각 LSTM 3개 처리)

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 : 총 1420개 中 1412개 데이터 파일 학습데이터로 사용
(결측 데이터 5개)

검증 : 학습 데이터에서 제외한 3개 데이터 사용 ++ 학습에서
검증에 사용된 3개 데이터 사용

- 학습 : ~2022년 56번 (874개)
- 검증 : ~ 2024년 92번 (249개)
- 테스트 : ~2025년 89번 (126개)

학습 : 총 1257개 中 1249개 데이터 파일 학습데이터로 사용 (결측 데이터 5개)

검증 : 학습 데이터에서 제외한 3개 데이터 사용 + 학습에서 검증에 사용된 3개 데이터 사용

컬럼 정리 (+ MIET 학습 데이터 기본 설정)

궁내교

time	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	1.12	5.11	0.0

대곡교

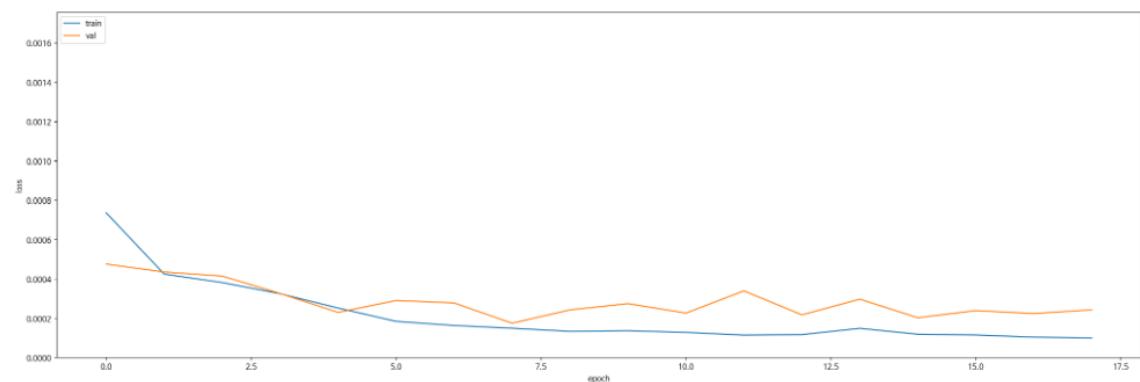
time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

0. seq_length = 12

num_of_layer = 12
학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

Train (19874 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0334	0.0312	0.0493	0.945846	0.954264	17.890869

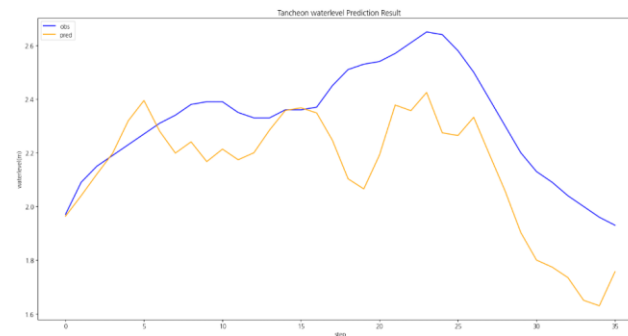
Valid (5882 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0389	0.0419	0.0538	0.938219	0.942474	0.141742

Test (4751 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0414	0.0477	0.0570	0.908699	0.929418	2.842335

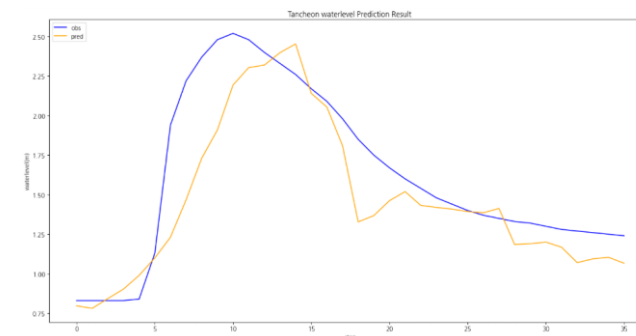
num_of_layer = 12

예측 코드 - 궁내교

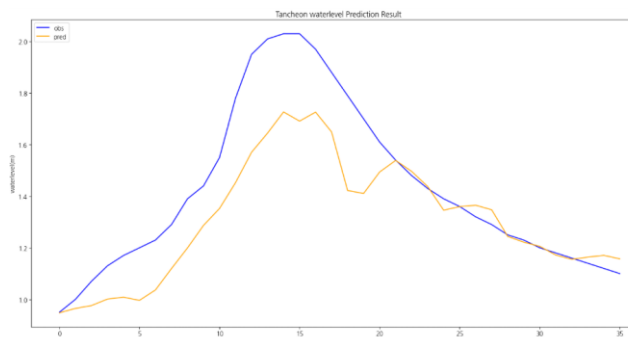
	궁내교
증설 전	testdata_220712 r2 : -0.3354817263001719 mse : 0.052989771533857255 mae : 0.19141196118460757 rmse : 0.2301950727836225
	testdata_230711 r2 : 0.7409383899519266 mse : 0.07249605282130708 mae : 0.20018566151460013 rmse : 0.2692509105301356
증설 후	testdata_250716 r2 : 0.6647419430513152 mse : 0.03363383329099639 mae : 0.13416874441835613 rmse : 0.1833952924450254
	testdata_250814 r2 : 0.7008753467118882 mse : 0.021088911233172896 mae : 0.1155578705999586 rmse : 0.1452202163377155



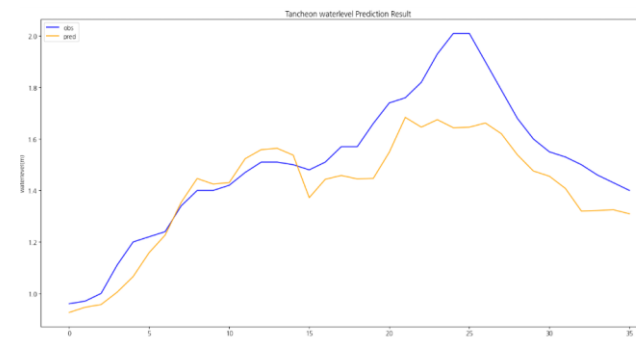
testdata_220712



testdata_230711



testdata_250716



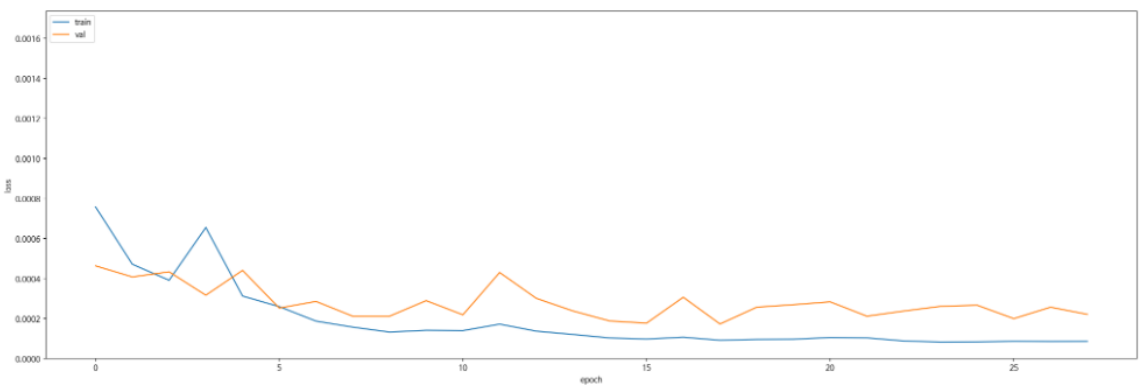
testdata_250814

1. seq_length = 18

num_of_layer = 18
학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

Train (19874 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0262	0.0252	0.0381	0.967542	0.968092	18.370273

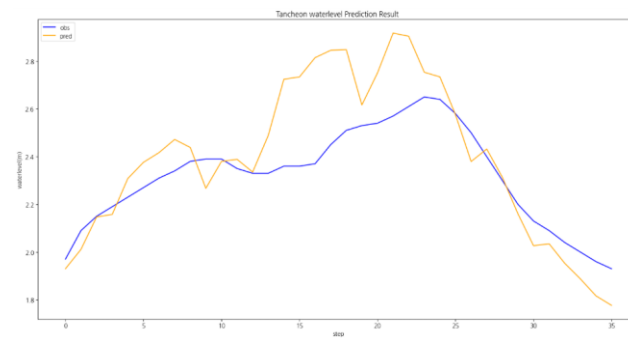
Valid (5882 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0390	0.0430	0.0535	0.938900	0.950641	2.229004

Test (4751 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0432	0.0504	0.0581	0.905261	0.935714	2.049240

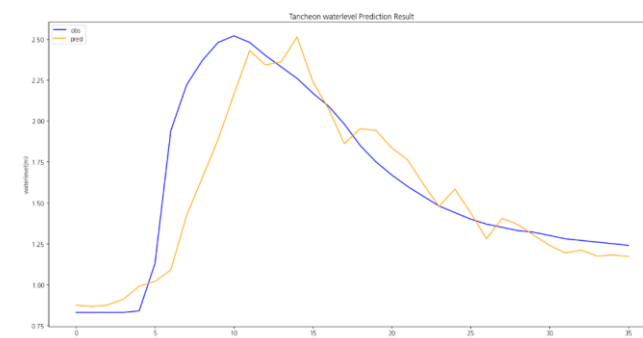
num_of_layer = 18

예측 코드 - 궁내교

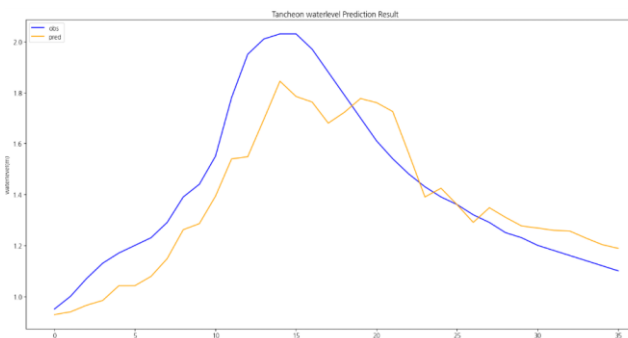
	궁내교
증설 전	testdata_220712 r2 : 0.14864938999199162 mse : 0.03378022583994122 mae : 0.13584673484166465 rmse : 0.18379397661496205
	testdata_230711 r2 : 0.7356497296198843 mse : 0.07397603667037891 mae : 0.1636988114648395 rmse : 0.27198536113250454
증설 후	testdata_250716 r2 : 0.7750344623088323 mse : 0.022569042664595026 mae : 0.12452420638667212 rmse : 0.15022996593421375
	testdata_250814 r2 : 0.7074602847267737 mse : 0.020624659384502603 mae : 0.10934795512093437 rmse : 0.14361288028760721



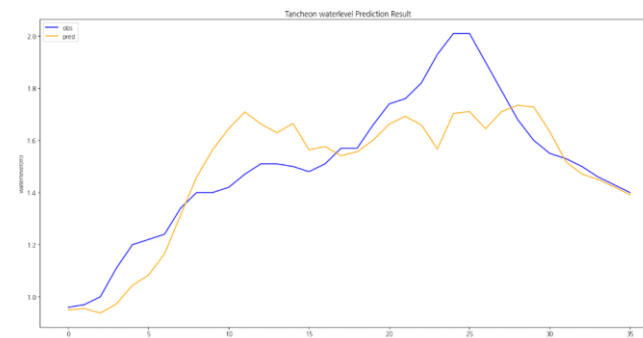
testdata_220712



testdata_230711



testdata_250716



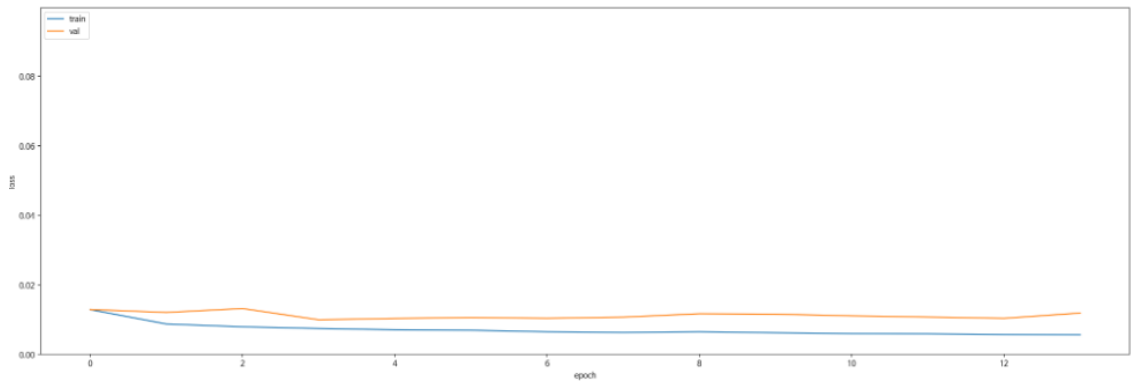
testdata_250814

**1-1. seq_length = 18
+ MAE**

num_of_layer = 18
loss_function = MAE
학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

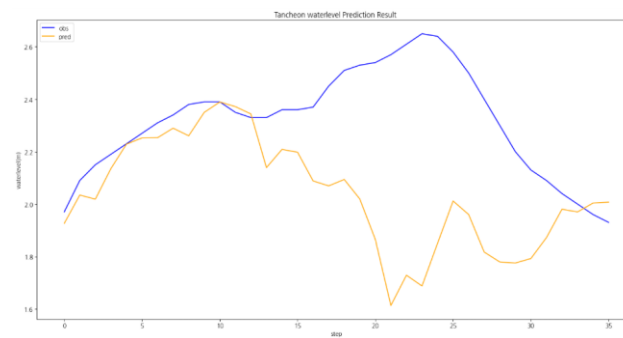
Train (19874 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0287	0.0259	0.0623	0.913279	0.920168	17.730663

Valid (5882 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0404	0.0435	0.0620	0.917917	0.930954	2.849581

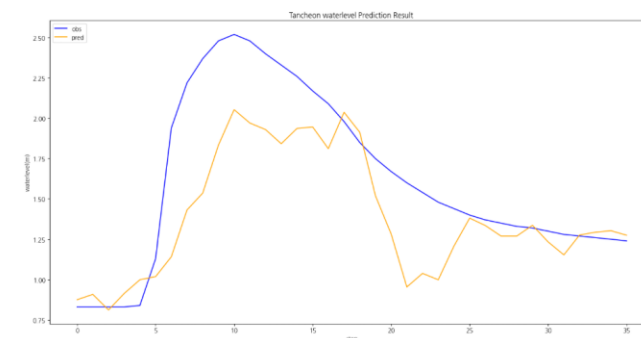
Test (4751 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0453	0.0525	0.0659	0.878156	0.905901	3.406047

num_of_layer = 18
 loss_function = MAE
 예측 코드 - 궁내교

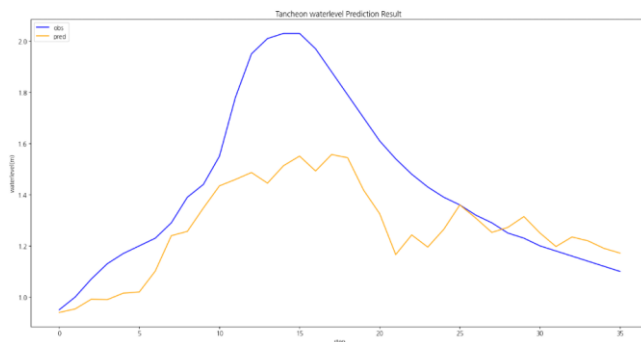
	궁내교
증설 전	testdata_220712 r2 : -3.2503557457995864 mse : 0.1686472944347232 mae : 0.2880982339382171 rmse : 0.4106668898690558
	testdata_230711 r2 : 0.5240755004511839 mse : 0.13318317465811524 mae : 0.262581952214241 rmse : 0.36494270051353983
증설 후	testdata_250716 r2 : 0.40984873150276957 mse : 0.05920528670290548 mae : 0.1827994661198722 rmse : 0.2433213650769399
	testdata_250814 r2 : -0.12258855290443704 mse : 0.07914483170591467 mae : 0.20277521332105003 rmse : 0.28132691251623027



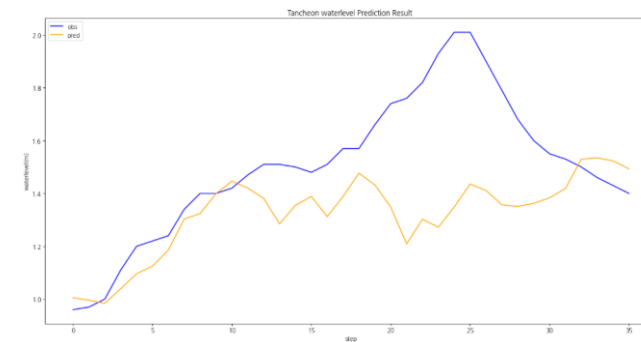
testdata_220712



testdata_230711



testdata_250716



testdata_250814

영향 없는 관측소 및 컬럼 빼기
seq_length = 18 성능이 더 좋음

결론

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- learning_rate = 0.0001
- Loss_function = MSE
- Seq_length = 36 (6시간)
- Num_of_layer = 12 (각 LSTM 3개 처리)

num_of_layer = 18 , 18(MAE) , 36

learning_rate = 0.0005 + num_of_layer = 18(MAE)